

1. // WAJP To Check Whether the Given Character is an Alphabet, Digit, or Special Character

```
import java.util.Scanner;

class CharacterCheck {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a character: ");

        char ch = scanner.next().charAt(0);

        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {

            System.out.println("Character is an Alphabet");

        }

        else if (ch >= '0' && ch <= '9') {

            System.out.println("Character is a Digit");

        }

        else {

            System.out.println("Character is a Special Character");

        }

    }

}
```

=====

2. // WAJP to Check Whether the Given Number is Positive or Negative

```
import java.util.Scanner;

class CheckNumberPositiveOrNegative {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int number = sc.nextInt();

        if (number >= 0) {
```

```
        System.out.println("Positive Number");

    } else {

        System.out.println("Negative Number");

    }

}

}
```

=====

3. // WAP: If the number is divisible by 3 print "Raja",
// if the number is divisible by 5 print "Rani",
// if the number is divisible by both 3 and 5 print "Raja weds Rani"

```
import java.util.Scanner;

class Divisible {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a number:");

        int n = sc.nextInt();

        if (n % 3 == 0 && n % 5 == 0) {

            System.out.println("Raja weds Rani");

        }

        else if (n % 3 == 0) {

            System.out.println("Raja");

        }

        else if (n % 5 == 0) {

            System.out.println("Rani");

        }

    }

}
```

=====

4. // WAJP to swap the values present in two variables

```
class Swapping {  
  
    public static void main(String[] args) {  
  
        int a = 10;  
  
        int b = 20;  
  
        System.out.println("Before swapping: a = " + a + ", b = " + b);  
  
        int temp = a;  
  
        a = b;  
  
        b = temp;  
  
        System.out.println("After swapping: a = " + a + ", b = " + b);  
  
    }  
}
```

=====

5. // WAJP to swap values without using a third variable

```
class SwappingWithoutThirdVariable {  
  
    public static void main(String[] args) {  
  
        int a = 10;  
  
        int b = 20;  
  
        System.out.println("Before swapping: a = " + a + ", b = " + b);  
  
        a = a + b;  
  
        b = a - b;  
  
        a = a - b;  
  
        System.out.println("After swapping: a = " + a + ", b = " + b);  
  
    }  
}
```

=====

6. // WAP to find the area of a Circle

```
import java.util.Scanner;

class AreaOfCircle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the radius of the circle: ");

        double radius = sc.nextDouble();

        double area = 3.14 * radius * radius;

        System.out.println("Area of the circle = " + area);

    }

}
```

7. // WAP to create a method which returns the area of a rectangle

```
import java.util.Scanner;

class AreaOfRectangle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter length: ");

        double length = sc.nextDouble();

        System.out.print("Enter breadth: ");

        double breadth = sc.nextDouble();

        double result = calculateArea(length, breadth);

        System.out.println("Area of the rectangle = " + result);

    }

    public static double calculateArea(double length, double breadth) {

        return length * breadth;

    }

}
```

8. // WAP to calculate Simple Interest

```
import java.util.Scanner;

class SimpleInterest {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the principal amount: ");

        double principal = sc.nextDouble();

        System.out.print("Enter the rate of interest: ");

        double rate = sc.nextDouble();

        System.out.print("Enter the time (in years): ");

        int time = sc.nextInt();

        double simpleInterest = (principal * rate * time) / 100;

        System.out.println("Simple Interest = " + simpleInterest);

    }

}
```

=====

9. // WAP to extract last digit from the given number and print it.

```
import java.util.Scanner;

class ExtractingLastDigit {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        while (num != 0) {

            int ld = num % 10;

            System.out.println(ld);

            num = num / 10;

        }

    }

}
```

```
    }  
    }  
}
```

=====

10. // WAP to remove the last digit from the given number and print the remaining number

```
import java.util.Scanner;  
  
class RemoveLastDigit {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("Enter the number");  
        int num = sc.nextInt();  
        int res = num / 10;  
        System.out.println(res);  
    }  
}
```

11. // WAP to find out the number of digits present in the given number.

```
import java.util.Scanner;  
  
class CountOfDigit {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("Enter a number:");  
        int num = sc.nextInt();  
        int count = 0;  
        while (num != 0) {  
            int ld = num % 10;  
            count++;  
            num = num / 10;  
        }  
    }  
}
```

```
    }  
  
    System.out.println(count);  
  
    }  
  
}
```

=====

12. // WAP to find out the sum of digits from the given number

```
import java.util.Scanner;  
  
class SumOfDigit {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Enter the number:");  
  
        int num = sc.nextInt();  
  
        int sum = 0;  
  
        while (num != 0) {  
  
            int ld = num % 10;  
  
            sum = sum + ld;  
  
            num = num / 10;  
  
        }  
  
        System.out.println(sum);  
  
    }  
  
}
```

=====

13. // WAP to identify the average of digits present in the given number

```
import java.util.Scanner;  
  
class AverageOfDigit {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);
```

```

System.out.println("Enter a number");

int num = sc.nextInt();

int count = 0;

int sum = 0;

while (num != 0) {

    int ld = num % 10;

    count++;

    sum = sum + ld;

    num = num / 10;

}

int res = sum / count;

System.out.println(res);

}

}

```

=====

14. // WJAP to identify the summation of only even digits

```

import java.util.Scanner;

class SumOfEvenDigit {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        int sum = 0;

        while (num != 0) {

            int ld = num % 10;

            if (ld % 2 == 0) {

                sum = sum + ld;

            }

            num = num / 10;

        }

        System.out.println(sum);

    }

}

```



```

    }

    num = num / 10;

}

System.out.println(sum);

}

}

```

=====

15. // WAP to identify the smallest digit in the given value/number

```

import java.util.Scanner;

class SmallestDigit {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        int temp = num;

        int small = num % 10;

        int ld = 0;

        while (num != 0) {

            ld = num % 10;

            if (ld < small) {

                small = ld;

            }

            num = num / 10;

        }

        System.out.println(small);

    }

}

```

16. // WAP to identify the largest digit in the given value/number

```
import java.util.Scanner;

class LargestDigit {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        int large = num % 10;

        int ld = 0;

        while (num != 0) {

            ld = num % 10;

            if (ld > large) {

                large = ld;

            }

            num = num / 10;

        }

        System.out.println(large);

    }

}
```

=====

17. // WAP to Reverse the given number.

```
import java.util.Scanner;

class ReverseNum1 {

    public static void main(String[] args) {

        System.out.println("Enter the number:");

        Scanner sc = new Scanner(System.in);

        int num = sc.nextInt();
```

```

int temp = num;

int rev = 0;

while (num != 0) {

    int ld = num % 10;

    rev = rev * 10 + ld;

    num = num / 10;

}

System.out.println(rev);

}

}

```

18. // WJJP to check whether the given number is a palindrome.

```

import java.util.Scanner;

class Palindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number:");

        int num = sc.nextInt();

        int temp = num;

        int rev = 0;

        while (num != 0) {

            int ld = num % 10;

            rev = rev * 10 + ld;

            num = num / 10;

        }

        if (rev == temp)

            System.out.println("Palindrome");
    }
}

```

```
        else

            System.out.println("Not Palindrome");

        }

    }

}
```

=====

19. // WAP to identify the second smallest digit in the given number

```
import java.util.Scanner;

class SecondSmallest {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a number");

        int num = sc.nextInt();

        int smallFirst = num % 10;

        int smallSecond = Integer.MAX_VALUE;

        while (num != 0) {

            int ld = num % 10;

            if (ld < smallFirst) {

                smallSecond = smallFirst;

                smallFirst = ld;

            }

            else if (smallSecond > ld && smallFirst != ld) {

                smallSecond = ld;

            }

        }

    }

}
```

=====

20. // WAP to identify the second largest digit in the given number

```
import java.util.Scanner;

class SecondLargest {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        int largeFirst = num % 10;

        int largeSecond = 0;

        while (num != 0) {

            int ld = num % 10;

            if (ld > largeFirst) {

                largeSecond = largeFirst;

                largeFirst = ld;

            }

            else if (largeSecond < ld && largeFirst != ld) {

                largeSecond = ld;

            }

            num = num / 10;

        }

        System.out.println(largeFirst);

        System.out.println(largeSecond);

    }

}
```

=====

21. // WAP to identify the largest of three numbers using decision-making statements.

```
import java.util.Scanner;
```

```
public class LargestNumFromThreeUsingDecisionMakingSt {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the first number");
```

```
        int a = sc.nextInt();
```

```
        System.out.println("Enter the second number");
```

```
        int b = sc.nextInt();
```

```
        System.out.println("Enter the third number");
```

```
        int c = sc.nextInt();
```

```
        if (a > b && a > c) {
```

```
            System.out.println("a is greater");
```

```
        }
```

```
        else if (b > a && b > c) {
```

```
            System.out.println("b is greater");
```

```
        }
```

```
        else {
```

```
            System.out.println("c is greater");
```

```
        }
```

```
    }
```

```
}
```

```
=====
```

```
22. // WAP to identify the largest of three numbers without using decision-making statements
```

```
    // (using conditional operator).
```

```
import java.util.Scanner;
```

```
public class LargestNumFromThreeUsingConditionalSta {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```

System.out.println("Enter the first number");

int a = sc.nextInt();

System.out.println("Enter the second number");

int b = sc.nextInt();

System.out.println("Enter the third number");

int c = sc.nextInt();

String res = (a > b && a > c) ? "a is greater"
           : (b > a && b > c) ? "b is greater"
           : "c is greater";

System.out.println(res);

}

}

```

=====

23. // WAP to identify the factors of the given number

```

import java.util.Scanner;

public class FactorOfNumber {

    public static void main(String[] args) {

        System.out.println("Enter the number");

        Scanner sc = new Scanner(System.in);

        int num = sc.nextInt();

        for (int i = 1; i <= num; i++) {

            if (num % i == 0) {

                System.out.println(i + " factor of this number");

            }

        }

    }

}

```

24. // WAP to identify the sum of factors of a given number.

```
import java.util.Scanner;

public class SumOfFactor {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        int sum = 0;

        for (int i = 1; i <= num; i++) {

            if (num % i == 0) {

                sum = sum + i;

            }

        }

        System.out.println("Factors sum is: " + sum);

    }

}
```

=====

25. // WAP to find the average of factors of the given number.

```
import java.util.Scanner;

public class AverageOfFactor {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        int sum = 0;

        int count = 0;
```



```

        for (int i = 1; i <= num; i++) {

            if (num % i == 0) {

                sum = sum + i;

                count++;

            }

        }

        int res = sum / count;

        System.out.println("Factors average is: " + res);

    }

}

```

=====

26. // WAP to identify the factorial of a number.

```

import java.util.Scanner;

public class Factorial {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the number");

        int num = sc.nextInt();

        int fact = 1;

        for (int i = 1; i <= num; i++) {

            fact = fact * i;

        }

        System.out.println(fact);

    }

}

```

=====

27. // WAP to check whether the given number is a perfect number.

// Note: A perfect number means the sum of factors excluding the given number

// must be equal to the given number.

```
import java.util.Scanner;
```

```
public class PerfectNum {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the number");
```

```
        int num = sc.nextInt();
```

```
        int sum = 0;
```

```
        for (int i = 1; i < num; i++) {
```

```
            if (num % i == 0) {
```

```
                sum = sum + i;
```

```
            }
```

```
        }
```

```
        if (sum == num)
```

```
            System.out.println("Perfect number");
```

```
        else
```

```
            System.out.println("Not perfect");
```

```
    }
```

```
}
```

=====

28. // WAP to check whether the given number is a strong number or not.

// Note: A strong number means the summation of each digit's factorial

// should be equal to the given number.

```
import java.util.Scanner;
```

```
public class StrongNum {
```

```
    public static void main(String[] args) {
```

```

Scanner sc = new Scanner(System.in);

System.out.println("Enter the number");

int num = sc.nextInt();

int temp = num;

int sum = 0;

while (num != 0) {

    int fact = 1;

    int ld = num % 10;

    for (int i = 1; i <= ld; i++) {

        fact = fact * i;

    }

    sum = sum + fact;

    num = num / 10;

}

if (sum == temp)

    System.out.println(temp + " is a Strong Number " + sum);

else

    System.out.println(temp + " is a NOT Strong Number " + sum);

}

}

```

=====

29. // WAP to check whether the given number is a prime number or not

// Note: A number which is divisible by 1 and itself is known as a prime number.

```

import java.util.Scanner;

public class PrimeNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
    }
}

```

```

System.out.println("Enter the number");

int num = sc.nextInt();

int count = 0;

for (int i = 1; i <= num; i++) {

    if (num % i == 0) {

        count++;

    }

}

if (count == 2) {

    System.out.println("Number is prime");

} else {

    System.out.println("Number is not prime");

}

}

```

=====

30. // WAP to identify the perfect numbers from the given range

```

import java.util.Scanner;

public class PerfectNumberInGivenRange {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a starting number: ");

        int start = sc.nextInt();

        System.out.print("Enter an ending number: ");

        int end = sc.nextInt();

        for (int num = start; num <= end; num++) {

            int sum = 0;

```

```

        for (int i = 1; i <= num / 2; i++) {
            if (num % i == 0) {
                sum = sum + i;
            }
        }

        if (num == sum) {
            System.out.println(num + " is a Perfect Number");
        }
    }
}

```

=====

31. // WAP to identify prime numbers from the given range

```

import java.util.Scanner;

class PrimeNumInGivenRange {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a starting number: ");
        int start = sc.nextInt();

        System.out.print("Enter an ending number: ");
        int end = sc.nextInt();

        for (int num = start; num <= end; num++) {
            int count = 0;

            for (int i = 2; i <= num / 2; i++) {
                if (num % i == 0) {
                    count++;
                    break;
                }
            }
        }
    }
}

```

```

    }

}

if (count == 0 && num > 1) { // Exclude 1

    System.out.println(num + " is a prime number");

}

}

}

}

```

=====

32. // WAP to check whether the given number is a perfect square or not

```

import java.util.Scanner;

class PerfectSquare {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int num = sc.nextInt();

        int sqrt = 0;

        int res = 0;

        for (int i = 1; i * i <= num; i++) {

            if (i * i == num) {

                sqrt = i;

                res = i * i;

                break;

            }

        }

        if (num == res) {

            System.out.println(num + " is a perfect square of " + sqrt);

```

```

    } else {

        System.out.println(num + " is not a perfect square");

    }

}

}

```

```

=====

```

33. // WAP to identify whether the given number is an Armstrong number or not.

// Note: The sum of the power of each digit based on the digit count

// must be equal to the given number.

```
import java.util.Scanner;
```

```
public class Armstrong {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter a number:");
```

```
        int num = sc.nextInt();
```

```
        int temp = num;
```

```
        int temp1 = temp;
```

```
        int count = 0;
```

```
        int sum = 0;
```

```
        // Step 1: Count the digits
```

```
        while (num != 0) {
```

```
            count++;
```

```
            num = num / 10;
```

```
        }
```

```
        // Step 2: Extract each digit and raise it to the power of count
```

```
        while (temp != 0) {
```

```
            int ld = temp % 10;
```

```
int power = 1;

for (int i = 1; i <= count; i++) {

    power = power * ld;

}

sum = sum + power;

temp = temp / 10;

}

// Step 3: Check whether the number is equal to the sum

if (sum == temp1) {

    System.out.println(temp1 + " is an Armstrong number");

} else {

    System.out.println(temp1 + " is not an Armstrong number");

}

}

}
```